

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE AI 2026	Date:	18-08-2026

Code of Conduct

I ___ Y SRAVAN KUMAR (192472289) ___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ SRAVAN _____

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	3	3	3	3	3	15	75
2	3	3	3	3	3	15	
3	3	3	3	3	3	15	
4	3	3	3	3	3	15	
5	3	3	3	3	3	15	

Faculty In-charge	Course Coordinator	Program Director
Signature: _Dr.T.Kumaragurubaran_____	Signature: _____	Signature: _____
Date: ____18-08-2026_____	Date: _____	Date: _____

1. Constituency Parsing vs Dependency Parsing

Introduction

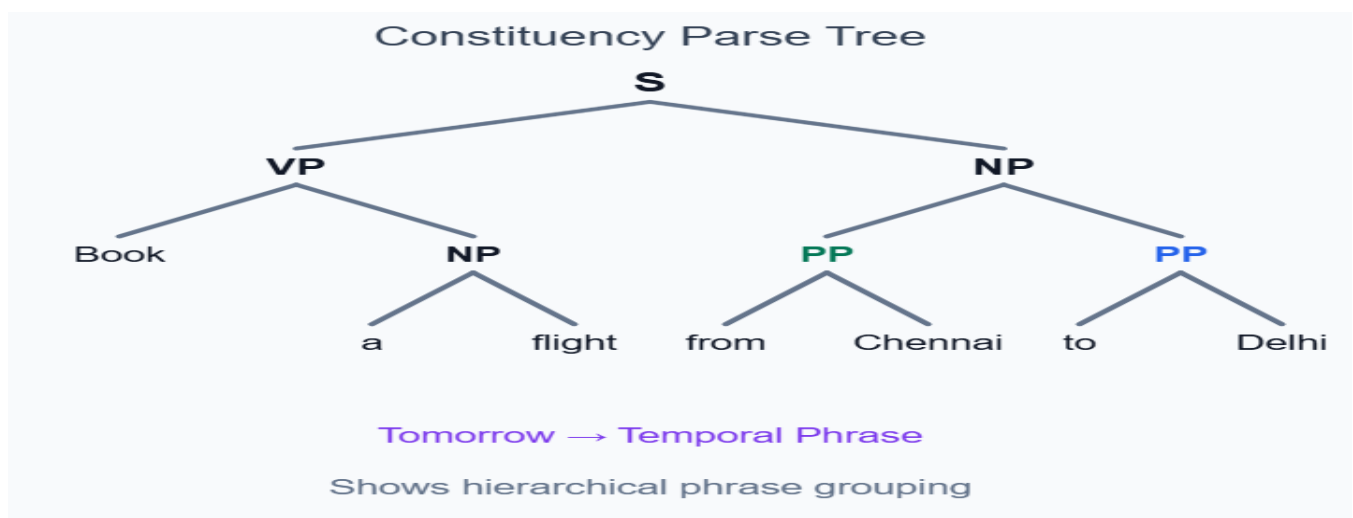
A conversational AI system must understand the grammatical relationships between words in user queries. The two common parsing methods are Constituency Parsing and Dependency Parsing. While both analyze sentence structure, they represent language in different ways.

Example Sentence

“Book a flight from Chennai to Delhi tomorrow.”

Constituency Parsing (Phrase Structure)

Constituency parsing groups words into phrases such as Noun Phrase (NP), Verb Phrase (VP), and Prepositional Phrase (PP).

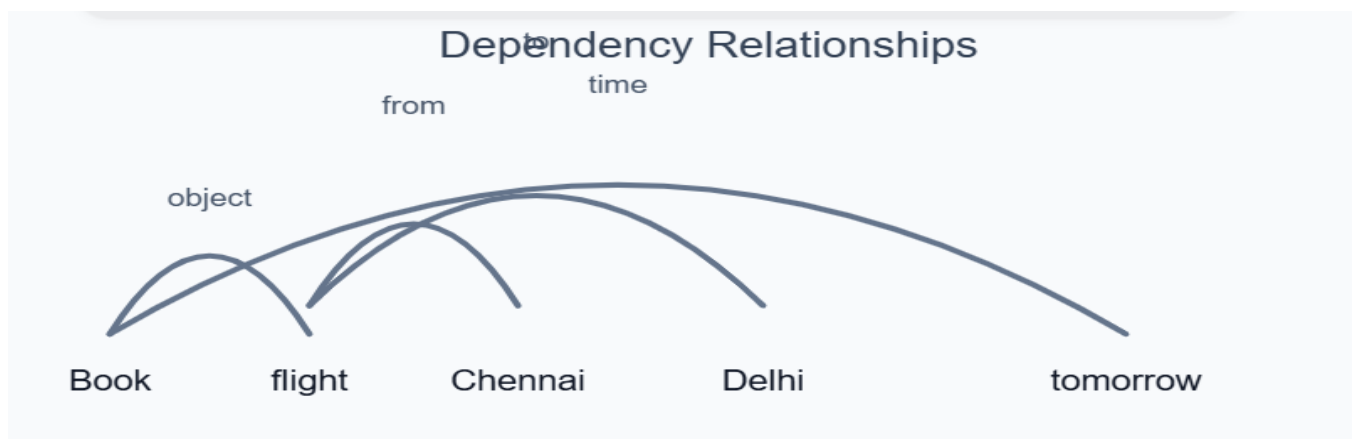


What it captures:

- Sentence hierarchy
- Phrase boundaries
- NP, VP, PP relationships

Dependency Parsing (Word-to-Word)

Dependency parsing connects words directly based on grammatical dependency.



What it captures:

- Direct grammatical relationships
- Subject, object, modifier
- Head-dependent connections

Difference Between the Two

Feature	Constituency Parsing	Dependency Parsing
Representation	Phrase structure	Word-to-word relations
Focus	Hierarchical phrases	Grammatical dependencies
Output	Parse tree	Dependency graph
Best for	Syntax analysis	Information extraction
Conversational NLP	Moderate	Excellent

Which is More Suitable?

Dependency Parsing is more suitable for conversational AI because it directly identifies meaningful relationships between words.

Example extraction:

Relationship Value

Action	Book
Object	Flight
Source	Chennai
Destination	Delhi
Time	Tomorrow

Justification: Dependency parsing is better for extracting entities, intents, locations, and actions from conversational text, making it ideal for chatbots and virtual assistants.

2. Top-Down Parsing vs Earley Parsing

Problem Statement

The system processes incomplete queries such as:

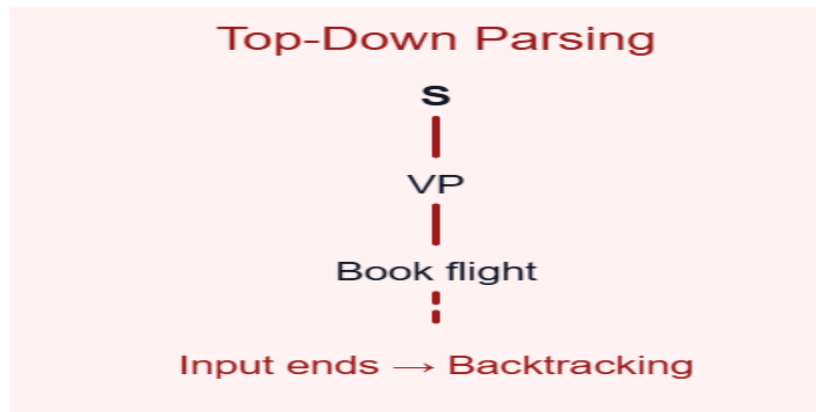
- “Best hotels near...”
- “Book a flight from Chennai to...”

These are partial sentences, making conventional top-down parsing inefficient.

Top-Down Parsing

Top-down parsing begins from the start symbol and expands grammar rules.

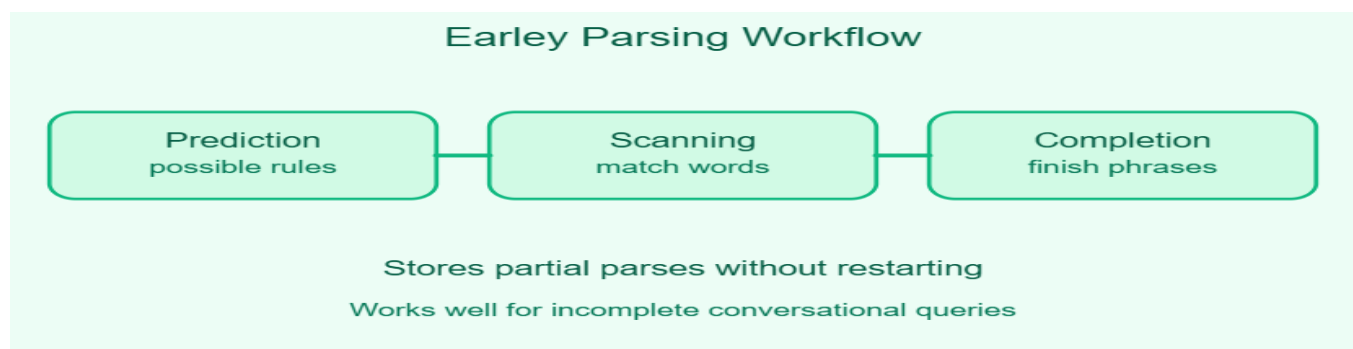
Example



Problem: When the input stops midway, the parser repeatedly backtracks and tries alternative grammar rules.

Earley Parsing

Earley parsing maintains partial parse states while reading the input.



Instead of restarting, it stores valid partial structures in a chart, making it efficient for autocomplete and conversational systems.

Comparative Analysis

Feature	Top-Down Parsing	Earley Parsing
Parsing Method	Expands grammar from root	Chart-based parsing
Backtracking	High	Very low
Ambiguity Handling	Poor	Excellent
Partial Input	Limited	Strong support
Long Queries	Slower	Efficient
Real-Time Chatbots	Less suitable	Highly suitable

Why Earley Parsing is Better

For queries like:

“Book a flight from Chennai to...”

Earley parsing can already identify:

- Intent: Book Flight
- Source: Chennai
- Destination: Waiting for input

without failing or restarting the parse.

3. CFG vs PCFG vs Neural Constituency Parsers

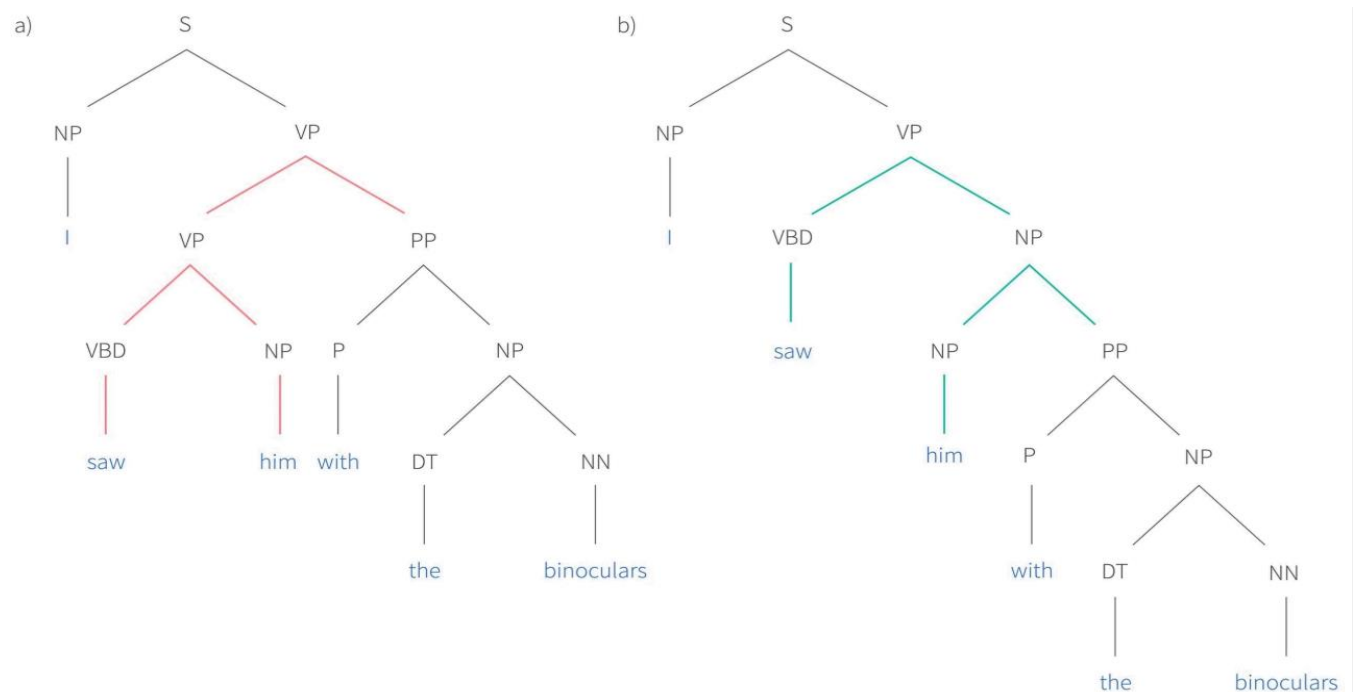
A grammar-checking system needs to identify the correct structure when a sentence has **more than one possible interpretation**.

For example:

“I saw the man with a telescope.”

This can mean:

1. I used a telescope to see the man.
2. The man had a telescope.



Comparison

Approach	How it represents ambiguity	How it resolves ambiguity
CFG	Produces multiple valid parse trees	Does not inherently choose the best one
PCFG	Assigns probabilities to grammar rules and parse trees	Selects the most probable parse
Neural Constituency Parser	Learns complex syntactic patterns from training data	Uses learned representations to predict the most likely structure

CFG

CFG uses rules such as:

$S \rightarrow NP VP$

$VP \rightarrow V NP PP$

$NP \rightarrow NP PP$

It can generate both interpretations, but it has **no probability mechanism** to decide which one is more likely.

PCFG

PCFG adds probabilities:

$VP \rightarrow V NP PP \quad 0.6$

$VP \rightarrow V NP \quad 0.4$

It can compare alternative parse trees and choose the one with the **highest probability**.

Neural Constituency Parser

A neural parser learns syntactic patterns from a large collection of annotated sentences. It can capture more complex language patterns and usually provides **higher practical accuracy**, especially on natural and conversational text.

Which is best?

For a grammar-checking system requiring a **balance between interpretability and practical accuracy**, **PCFG is a strong choice**.

- **CFG** $\rightarrow$ highly interpretable but cannot resolve ambiguity effectively.
- **PCFG** $\rightarrow$ interpretable because it still uses grammar rules and probabilities, while providing better ambiguity resolution.
- **Neural parser** $\rightarrow$ often more accurate, but its decisions are less transparent and require more training data and computational resources.

Conclusion

PCFG provides the best balance between interpretability and practical accuracy because it retains explicit grammar rules while using probabilities to resolve structural ambiguity. Neural constituency parsers may provide higher accuracy, but they are less interpretable.

4. Feature Structures vs Traditional CFG

A multilingual NLP system needs to handle grammatical information such as:

- **Number** – singular/plural
- **Gender** – masculine/feminine/neuter
- **Tense** – past/present/future
- **Person** – first/second/third

Traditional CFG mainly represents **syntactic structure**, but these grammatical features are not naturally encoded.

Feature Structures and Unification 1

Compare and combine feature structures:

he buys



Example

Consider:

“The boy plays.”

The subject is:

Number = Singular

Person = Third

Therefore, the verb should also agree:

plays → Singular, Third Person

But:

“The boy play.”

has an agreement problem.

How Feature Structures Extend CFG

Feature structures attach additional information to grammatical constituents.

For example:

NP

- |— Number = Singular
- |— Person = Third
- |— Gender = Masculine

The corresponding verb can be required to have matching features:

VP

- |— Tense = Present
- |— Number = Singular
- |— Person = Third

The parser can then check whether the features are compatible.

Traditional CFG vs Feature Structures

Feature	Traditional CFG	Feature Structures
Basic syntax	✓	✓
Number	Limited	✓
Gender	Limited	✓
Person	Limited	✓
Tense	Limited	✓
Agreement constraints	Difficult	Effective
Multilingual grammar	More rules required	More flexible

Why Useful in Multilingual NLP?

Different languages have different grammatical agreement requirements.

For example, a multilingual system may need to track:

Person

↓

Number

↓

Gender

↓

Tense

Feature structures allow these properties to travel with grammatical constituents and be checked during parsing.

This reduces the need to create a separate rigid CFG rule for every possible combination.

5. Subcategorization Frames

Subcategorization describes **what types and how many arguments a verb requires** to form a valid sentence.

Examples

GIVE

give + object + recipient

↓

↓

book John

"Give the book to John."

SLEEP

sleep + no object

"Sleep."

PUT

put + object + location

↓ ↓ ↓
file file desk

"Put the file on the desk."

Picture / Diagram

<i>Subcategory</i>	<i>Example</i>
V[NP__] (intransitive)	<i>leave</i>
V[NP ____ NP] (transitive type 1)	<i>hit</i>
V[NP ____ {NP/CP}] (transitive type 2)	<i>ask</i>
V[NP ____ NP NP] (ditransitive type 1)	<i>spare</i>
V[NP ____ NP PP] (ditransitive type 2)	<i>put</i>
V[NP ____ NP {NP/PP}] (ditransitive type 3)	<i>give</i>
V[NP ____ NP {NP/PP/CP}] (ditransitive type 4)	<i>tell</i>

How It Helps

Verb Required Structure

give Verb + Object + Recipient

sleep Verb only

put Verb + Object + Location

For example, the system can recognize that:

“Put the file.”

is incomplete because **put** normally requires a location.

Importance

Subcategorization helps the voice assistant:

- **Syntactically:** determine whether the sentence structure is valid.
- **Semantically:** identify the role of each argument.
- **Command understanding:** determine **what object to act on, who receives it, and where it should go**